

Canonical Camera Space Transform — Label Space

Idea: Map all training data to a “canonical” camera with fixed focal length $f^c = 1000$ px.

CSTM_{label} transforms depth annotations:

$$D_c^* = \omega_d \cdot D^*, \quad \omega_d = \frac{f^c}{f}$$

If $f < f^c$: depth is *scaled up* (wide-angle cameras see closer).

If $f > f^c$: depth is *scaled down*.

At inference, **De-CSTM** reverses:

$$D = \frac{D_c}{\omega_d} = D_c \cdot \frac{f}{f^c}$$

